COLLECTION OF CTL DECISIONS

Standard:	**Sub clause:**	**Sheet n. 323**
IEC 61010-1:1990+A1:1992+A2:1995	6.7	**Page 1(1)**
Subject:	**Key words:**	**Provisional decision taken by ETF3 and confirmed by CTL at its 38th meeting, in Toronto**
Clearances and Creepage distances	- Clearance - Creepage - Pulses	

Question:

Do creepage and clearance distances have to be calculated from the peak value of voltage spikes occurring between parts even when the parts in question are subject only to very short duration pulses occurring at defined times e.g. switch on?

Decision:

Clearance distances must be compliant with the value for the highest voltage present on the parts concerned, creepages distances must not be less than clearance values, therefore no reduction in values is allowed.

Where the distances are less than required values they should be shorted out and measurements carried out in accordance with Clause 6.3.2. 'Values in single fault condition' to determine whether a hazard exists. If no hazard exists then the device is compliant with the requirement.

Explanation:

Clearances are liable to break down under excessive voltage and therefore cannot be reduced from the stated values. However for very short duration pulses the energy transmitted to accessible parts may be within the limit allowed by IEC61010-1 and the clearance is not then essential as a means of protective isolation.